Telco Customer Churn

Predictive Modeling & A/B Testing

Statistical Methods for Machine Learning

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IBM Telco Customer Churn Dataset | Logistic Regression · Random Forest · SVM

Agenda

01
Dataset Overview
IBM Telco — 7,043 customers, 21 features

02
Research Question
Can ML outperform random targeting?

03
Exploratory Data Analysis
Churn patterns across key segments

04
Preprocessing
Encoding, scaling & train/test split

05
ML Models
LR, Random Forest, SVM — training & evaluation

06
A/B Testing
Simulated retention campaign comparison

Dataset Overview

IBM Telco Customer Churn · Kaggle

7,043	**21**	**26.5%**	**Binary**
Customers	Features	Churn Rate	Target Variable

Key Feature Groups

Demographics

Gender, senior citizen, partner, dependents

Services

Phone, internet, streaming, security, backup

Account Info

Tenure, contract type, billing, payment method

Financials

Monthly charges, total charges

Research Question

"Does targeting high-risk customers using machine learning reduce churn more effectively than a random retention strategy?"

Approach: Train classifiers → rank customers by churn risk → simulate A/B campaign → compare outcomes statistically

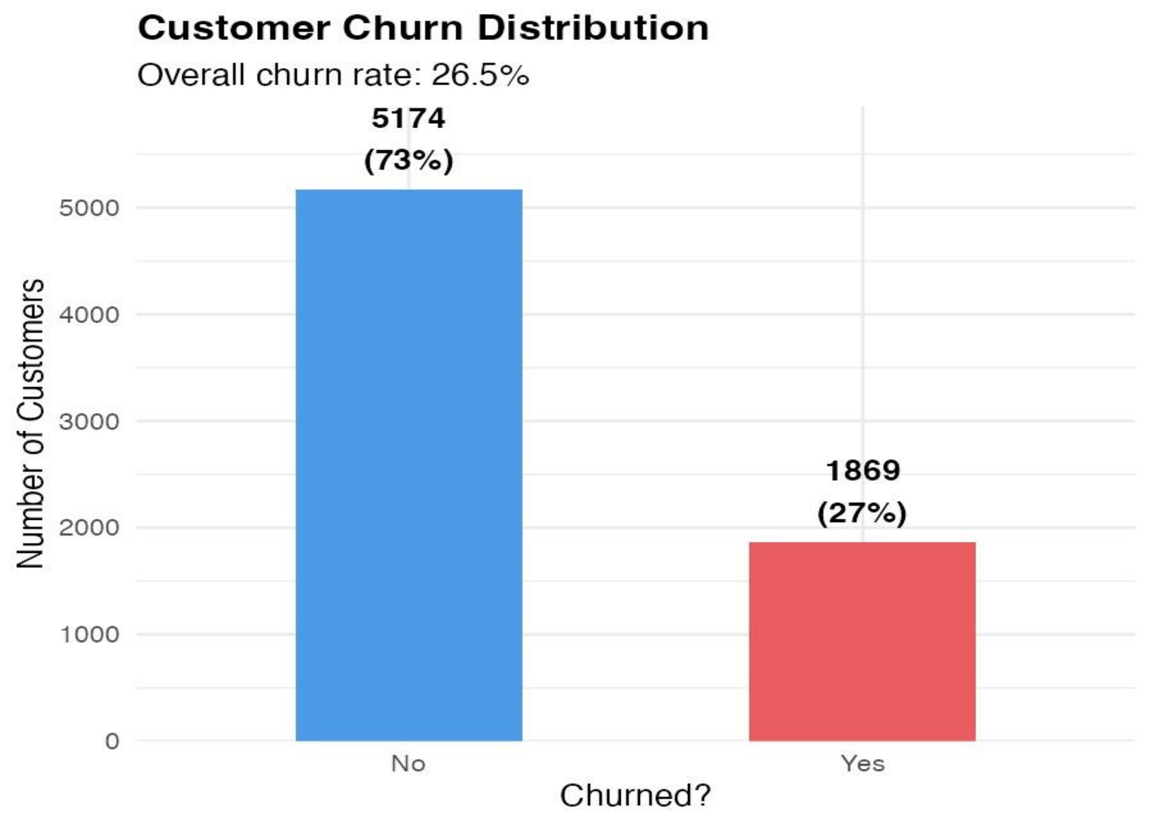
EDA · Overall Churn Rate

26.5%

of customers churned

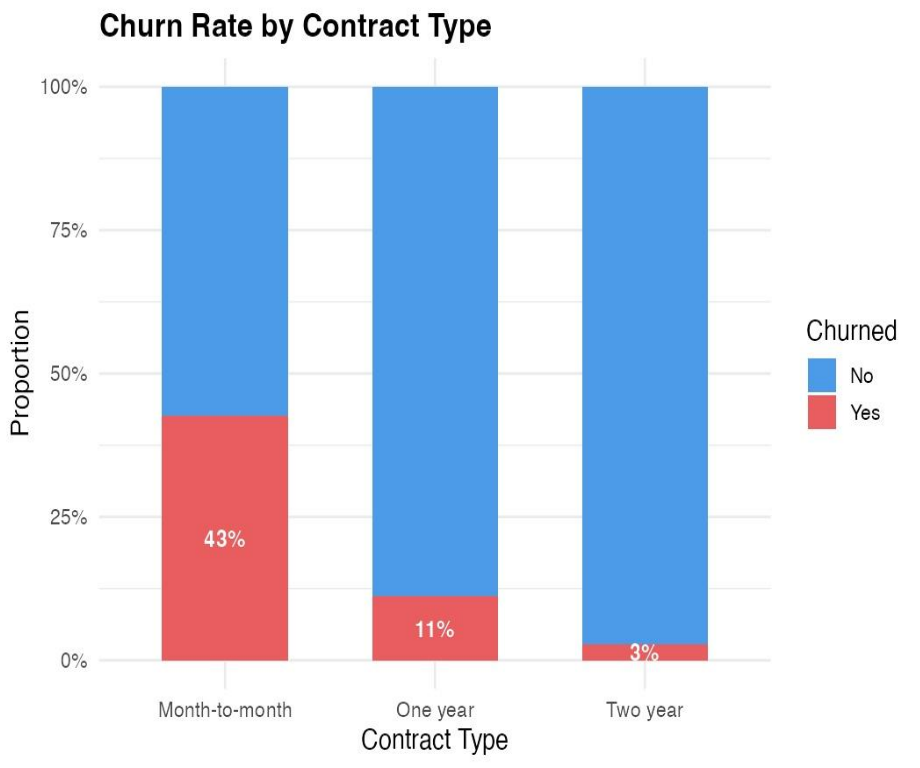
5,174 retained
1,869 churned

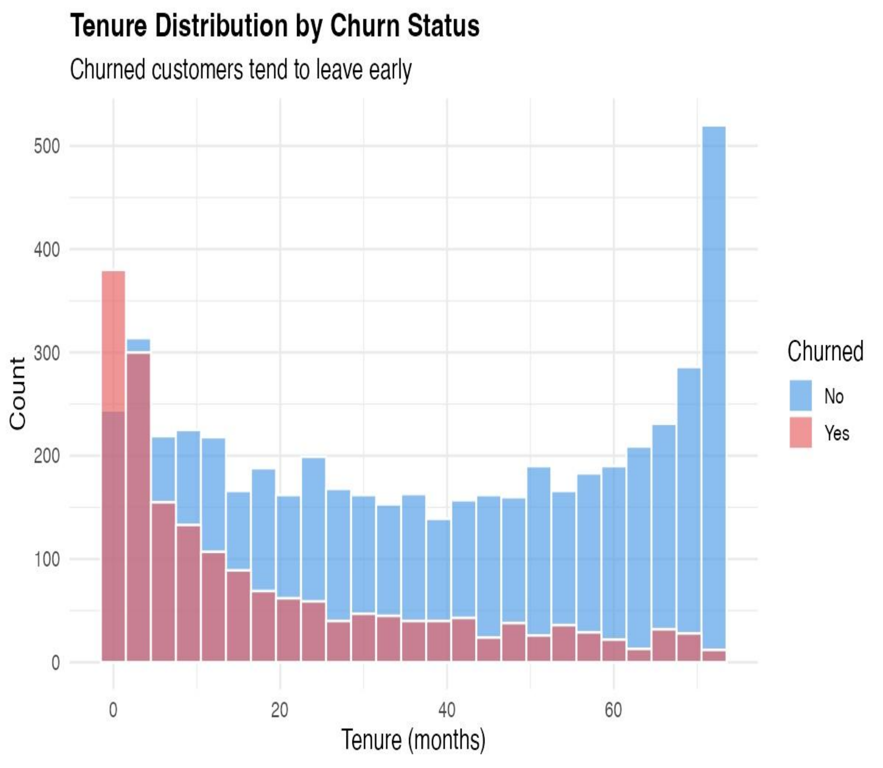
⚠️ Class imbalance — important for model evaluation



EDA · Contract Type & Tenure

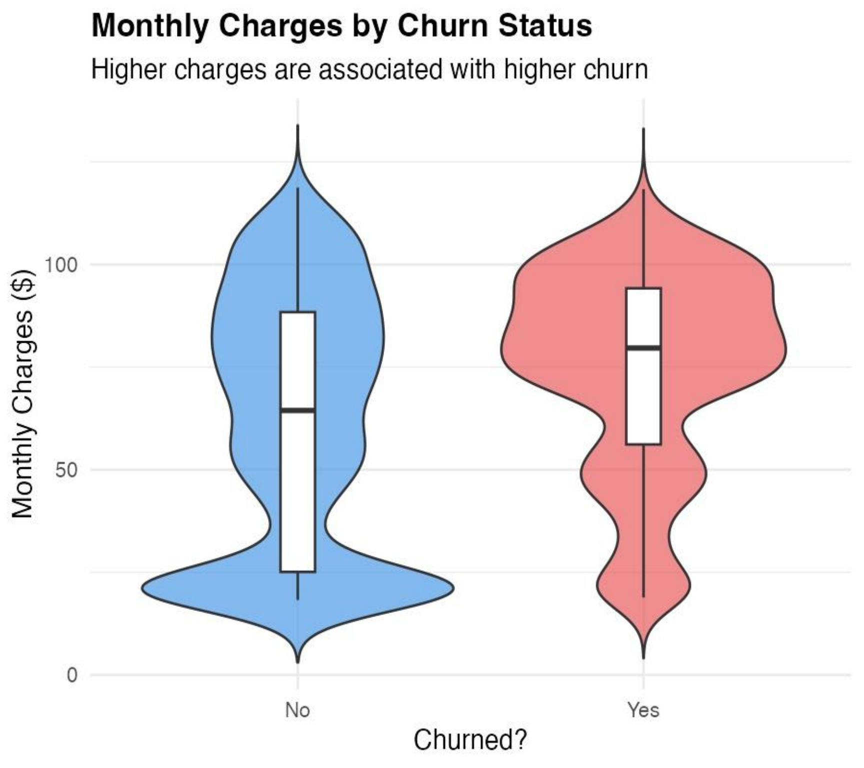
Contract type is the strongest predictor — month-to-month customers churn at 43%

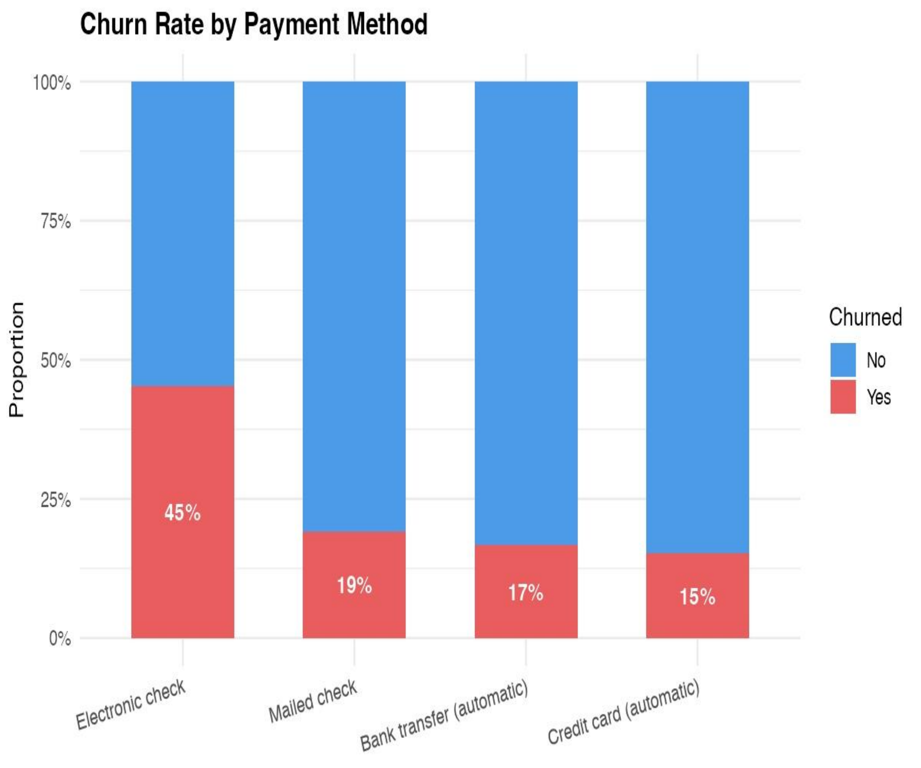




EDA · Monthly Charges & Payment Method

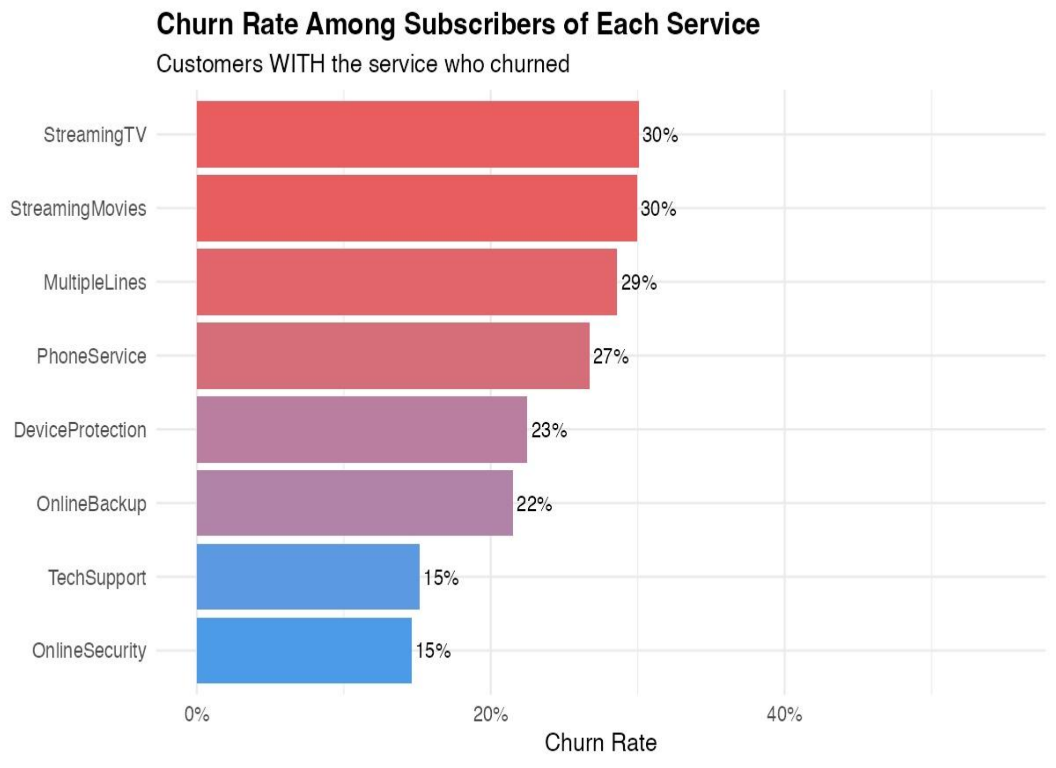
Higher bills and manual payment methods are associated with significantly higher churn





EDA · Services & Feature Correlations

TechSupport & OnlineSecurity subscribers show lower churn · TotalCharges is highly correlated with tenure



Preprocessing Pipeline

01 **Clean**

Remove customerID · Impute blank TotalCharges ⟶ 0 (new customers)

02 **Encode Binary**

Yes/No columns ⟶ 0/1 · Gender ⟶ 0/1

03 **One-Hot Encode**

Multi-level categoricals via model.matrix() · janitor cleans column names

04 **Stratified Split**

80% train / 20% test · Churn rate preserved in both sets

05 **Scale Features**

Z-score normalization · Fit on train only — prevents data leakage

Models · Three Classifiers

All trained via 5-fold cross-validation · Optimizing AUC-ROC · Evaluated on held-out test set

Logistic Regression

Baseline

Interpretable linear model. Estimates log-odds of churn as a linear combination of features. No tuning parameters — serves as the explainable baseline.

family = binomial

Random Forest

Ensemble

Ensemble of 500 decision trees. Each tree sees a random subset of features at each split. Aggregated predictions reduce variance and handle non-linearity.

ntree = 500 · tuned: mtry $\in \{3, 5, 7, 10\}$

SVM (RBF Kernel)

Margin-Based

Finds the maximum-margin hyperplane separating churners from retained customers. RBF kernel maps features into higher-dimensional space for non-linear boundaries.

tuned: $C \in \{0.1, 1, 10\}$ · $\sigma \in \{0.01, 0.05, 0.1\}$

Models · Cross-Validation Methodology

5-Fold Cross-Validation

Fold 1	Val	Train	Train	Train	Train
Fold 2	Train	Val	Train	Train	Train
Fold 3	Train	Train	Val	Train	Train
Fold 4	Train	Train	Train	Val	Train
Fold 5	Train	Train	Train	Train	Val

Why AUC-ROC?

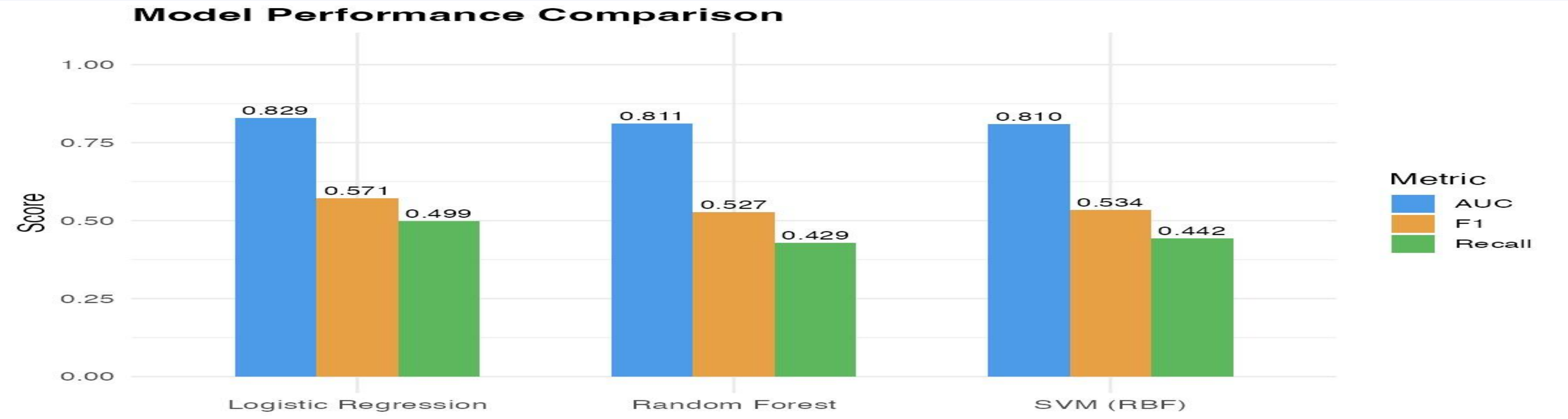
- Dataset is imbalanced (26.5% churn) — accuracy alone is misleading
- AUC measures the model's ability to rank churners above non-churners
- Threshold-free — evaluates performance across all classification cutoffs
- Standard metric for binary classification in imbalanced settings

Hyperparameter tuning (mtry for RF, C + σ for SVM) also performed within each fold to prevent leakage

Model Results · Performance Comparison

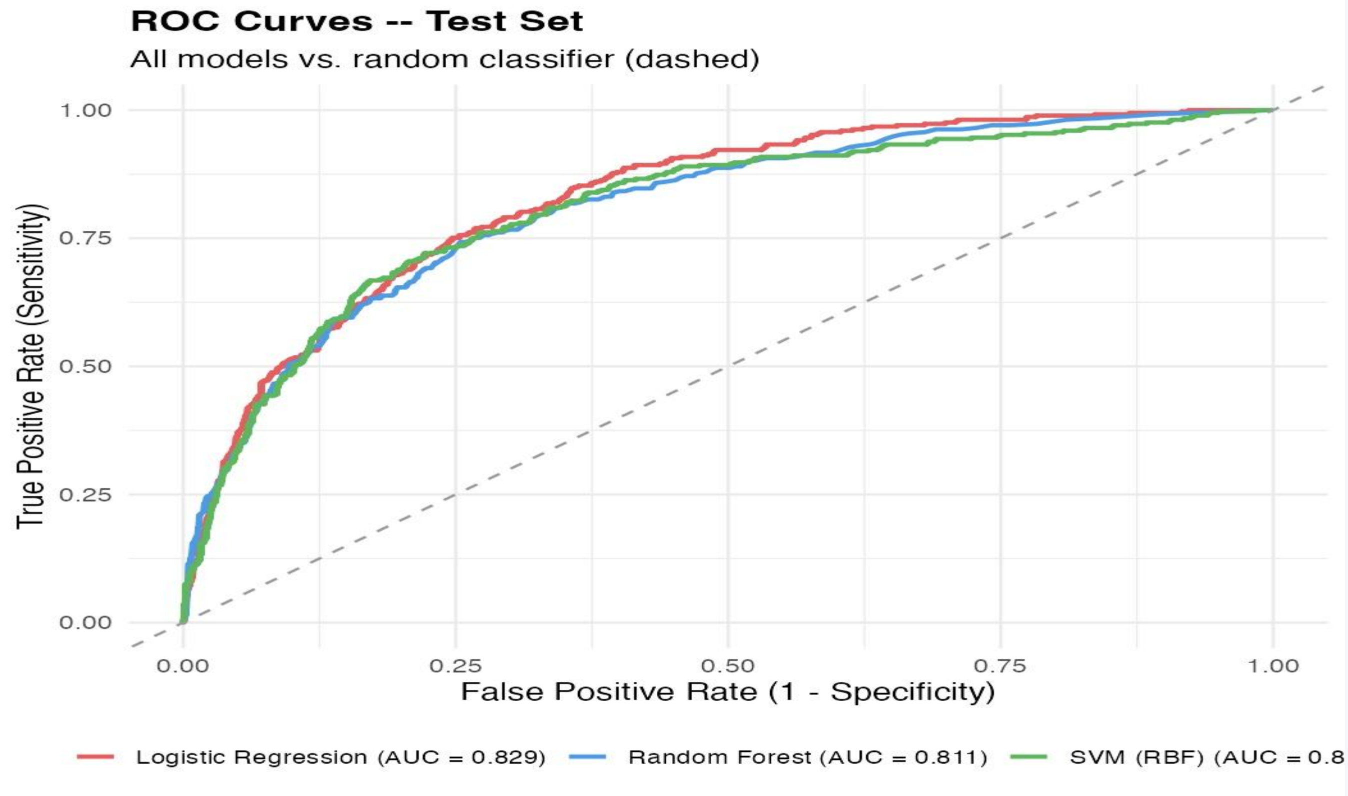
All three models achieve AUC ≈ 0.81–0.83 on the held-out test set

Model	AUC	Accuracy	Precision	F1	Recall
Logistic Regression	**0.829**	0.799	0.331	0.571	0.499
Random Forest	**0.811**	0.799	0.337	0.527	0.429
SVM (RBF)	**0.810**	0.799	0.327	0.534	0.442



Model Results · ROC Curves

Logistic Regression edges out the ensemble methods — all well above the random baseline



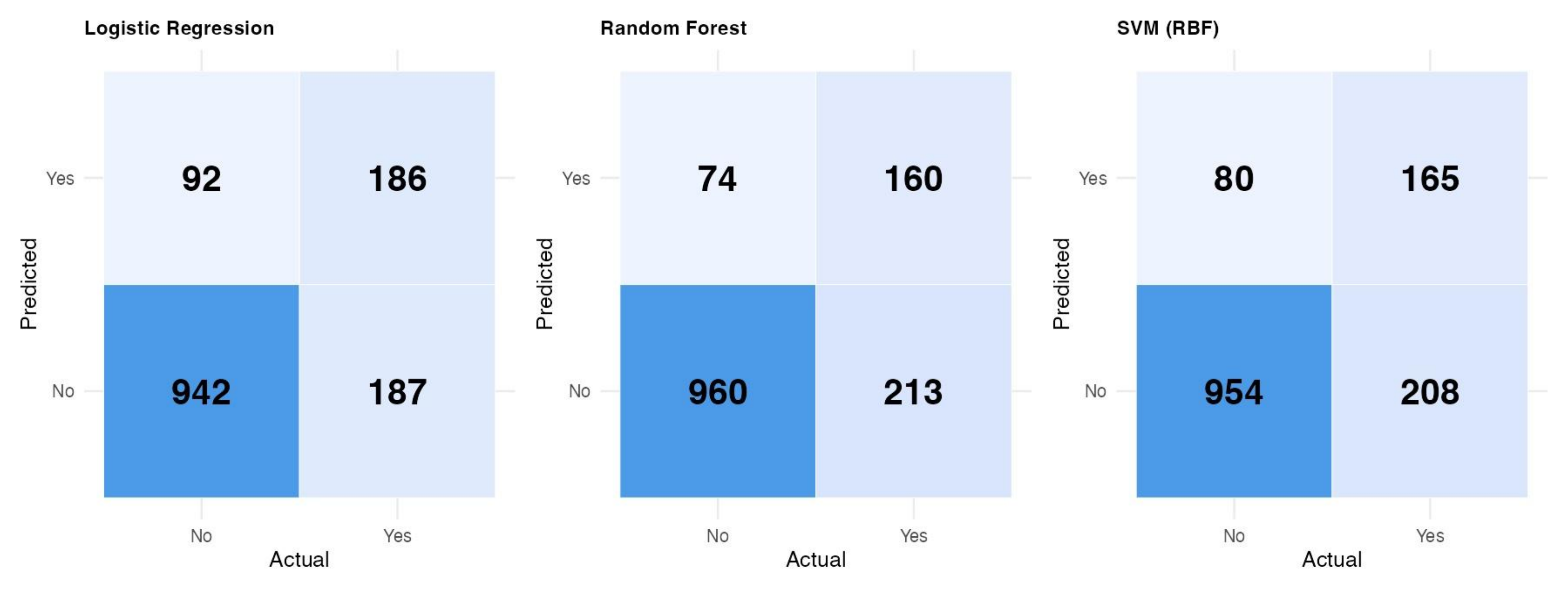
LR
AUC = 0.829

RF
AUC = 0.811

SVM
AUC = 0.810

Model Results · Confusion Matrices

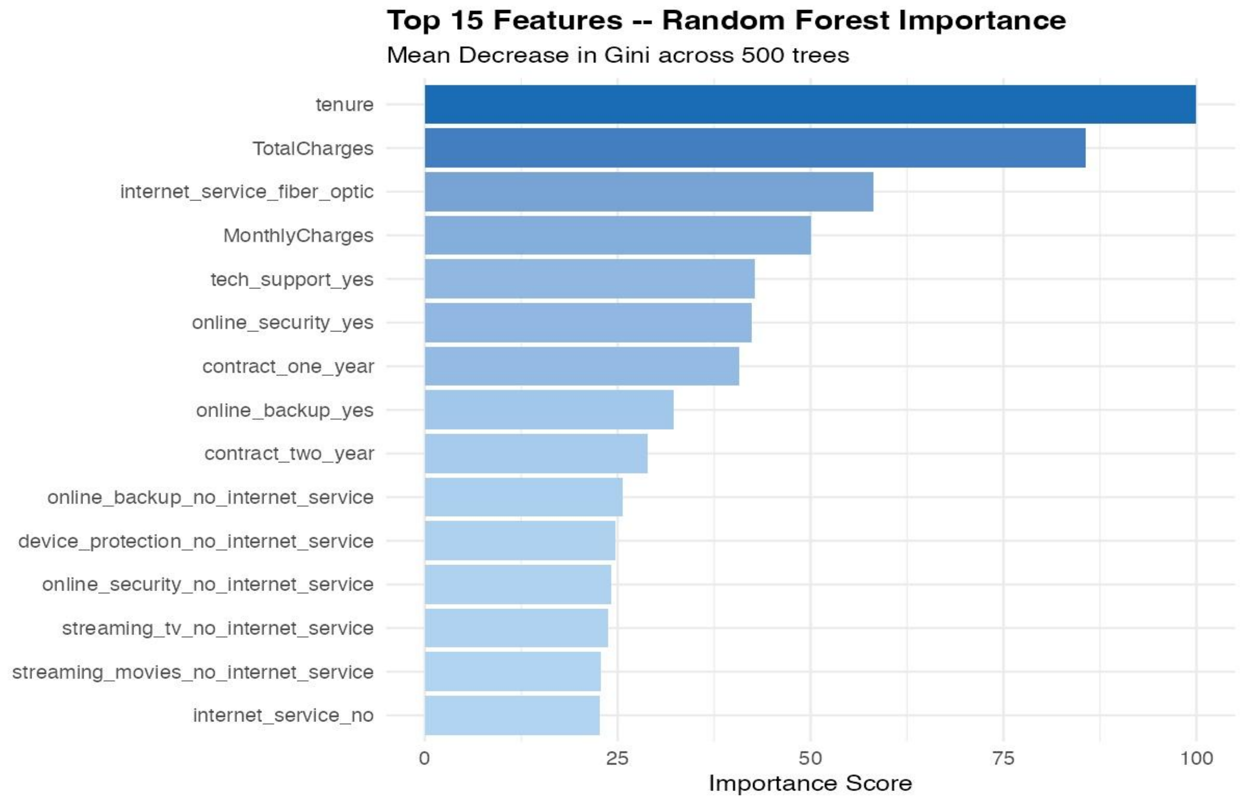
False negatives (missed churners) are the costlier error — retaining an at-risk customer is more valuable than a false alarm



All models predict 'No churn' conservatively — recall for churners is ~43–50%, leaving significant room for improvement

Model Results · Feature Importance

Random Forest — Mean Decrease in Gini across 500 trees



Tenure

Longer-tenured customers are far less likely to churn

Total Charges

Strongly correlated with tenure — captures lifetime value

Fiber Optic

Higher churn risk — often paired with high monthly charges

A/B Testing · Experimental Design

Simulated retention campaign using test set customers — model ranks by churn risk, random is the baseline

Group A — ML-Targeted

Top 30% of customers ranked by predicted churn probability from the best-performing model.

Contacted with a simulated retention offer.

VS

Group B — Random

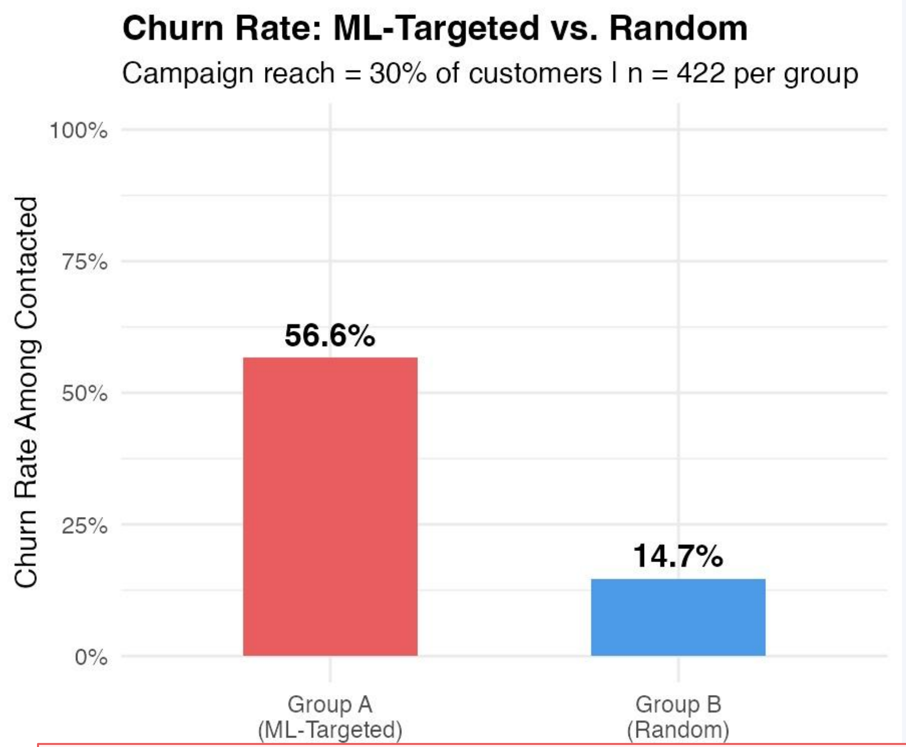
Same number of customers (30%) selected uniformly at random — no model involvement.

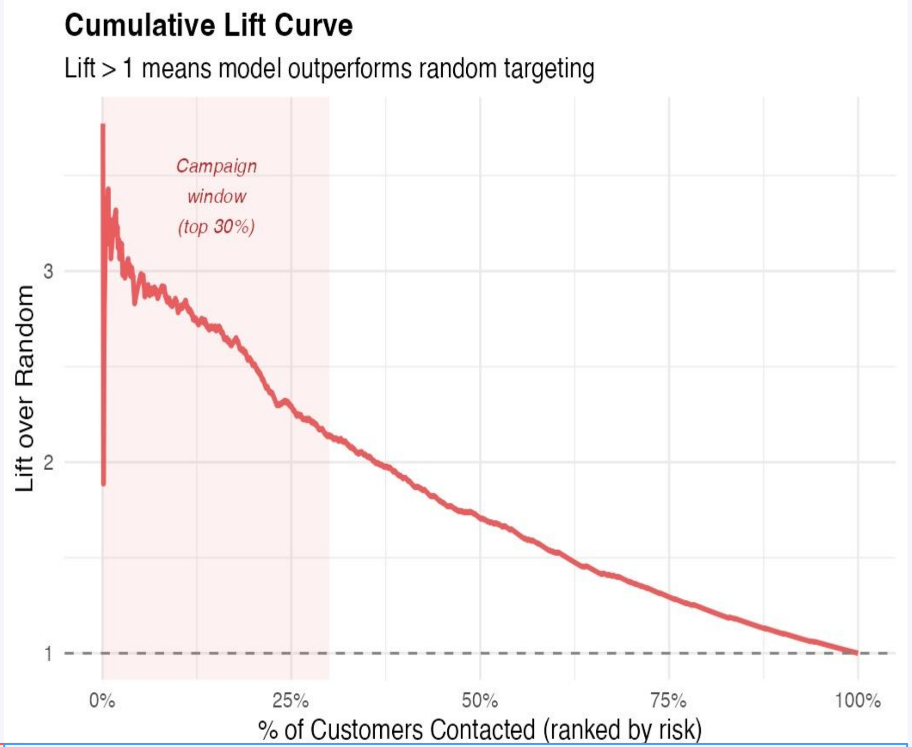
Baseline 'no model' strategy.

Outcome: Of those contacted, how many were true churners? · Tested with Z-test, Chi-square & Fisher's Exact

A/B Testing · Results

ML targeting identified 3.9× more true churners per customer contacted than random selection





56.6% vs 14.7% churn rate · $p < 0.001$ · Statistically significant

Lift ≈ 2.1× at 30% budget depth · Model consistently outperforms random

Conclusions & Takeaways

EDA	Contract type, tenure, and monthly charges are the strongest churn drivers — month-to-month customers churn at 43%
Modeling	Logistic Regression led with AUC = 0.829 — competitive with Random Forest and SVM, and more interpretable
Feature Importance	Tenure and Total Charges dominate — long-term customers are significantly less likely to leave
A/B Testing	ML targeting found 3.9× more true churners per contact vs. random — result is statistically significant ($p < 0.001$)
Business Value	At a 30% campaign budget, the model delivers ~2.1× lift — meaningful ROI improvement over random outreach